

These equations can be uncoupled by dividing the second one by α , multiplying the first one by β/α^2 , and subtracting, to obtain

$$\frac{d}{dx} \left(\frac{\beta}{\alpha} \right) + \left(\frac{\beta}{\alpha} \right)^2 = 1 + x^2. \quad (14)$$

Clearly, Mielnik's solution (7) is the general solution for this equation, introducing one parameter, γ . A second parameter, δ , will appear when we insert this solution into one of the equations (12) or (13) to find the complete biparametric solution. When these parameters acquire the values that make $\alpha \equiv 1$ we shall recover Mielnik's factorization of the simple harmonic oscillator, just as when $\gamma \rightarrow \infty$ in Mielnik's factorization we recover the original factorization (4).

Using Mielnik's solution (7) it is easy to calculate the general solution to eqs.(12,13), and to find the operators that factorize the SHO hamiltonian in terms of the product BB^* . Then, we may consider carrying out the inverse product B^*B to see where we are led to. Obviously, we shall not obtain a newer Hamiltonian, due to the factors α^{-1} and α in (11), which may be the reason why this factorization was not paid attention to in the past. What's more, it is obvious that this operator product will be lengthy and most easily won't give any new insight to the problem. Therefore, we shall consider here only the most simple solution to eqs.(12,13) to show that it leads to a new general equation for the SHO, lost in the trends of the SUSY factorization scheme.

III. NEW GENERALIZED HERMITE EQUATION

Let us now take the simplest solution to eq.(14), $\beta = \alpha x$. Introducing this solution into eq.(12), we get a Bernoulli equation for $\alpha(x)$

$$\alpha' + x\alpha^3 - x\alpha = 0. \quad (15)$$

This equation is easily integrated, giving α and β as

$$\alpha(x) = \frac{1}{\sqrt{1 + \delta e^{-x^2}}}, \quad \beta(x) = \frac{x}{\sqrt{1 + \delta e^{-x^2}}}. \quad (16)$$

To avoid singularities, we simply require that $0 \leq \delta < \infty$.

As we said before, the inverse operator product B^*B will not give us a new Hamiltonian.